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INTELLIGENT INTRUSION DETECTION SYSTEM USING FACE RECOGNITION AND CRYPTOGRAPHIC SECURITY

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA5113 - CRYPTOGRAPHY AND NETWORK SECURITY

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING

Submitted by

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SEPTEMBER 2026

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Intelligent Intrusion Detection System Using Face Recognition and Cryptographic Security” has been carried out by Kartheeka Repalle (192372289) under the supervision of the Project Guide and is submitted in partial fulfilment of the requirements for the current semester of the B.E. Computer Science Engineering (CSE-AI) programme at Saveetha Institute of Medical and Technical Sciences, Chennai.

The project presents an integrated intelligent surveillance prototype that combines camera-based motion detection and face recognition with cryptographic evidence protection, secure event logging, and authenticated intrusion notification.

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DECLARATION

I, Kartheeka Repalle (192372289) of the Computer Science Engineering (CSE-AI) programme, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Intelligent Intrusion Detection System Using Face Recognition and Cryptographic Security” is the result of our own bona fide efforts.

To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, software libraries, datasets, computational resources, and external tools used in the project have been appropriately acknowledged.

We further declare that the data, results, analysis, and findings presented in the final submission have not been fabricated, falsified, or intentionally manipulated. Any AI-assisted tools used during project planning, coding support, documentation, or visual preparation have been used as assistance and should be disclosed wherever required by institutional policy.

We confirm that all applicable ethical, academic, institutional, privacy, and professional requirements have been followed throughout the planning, implementation, testing, analysis, and documentation of the project.

Place: Chennai

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Signature of the Students with Names

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Kartheeka Repalle / 192372289	Module 1 lead; camera integration; motion detection; face detection/recognition workflow; system architecture; integration coordination.	Designed and implemented the camera-processing flow, face matching workflow, authorized/unknown decision logic, and integrated the system modules.	Prepared authorized vs unknown test cases; checked face-recognition outputs; analyzed threshold behaviour and end-to-end event flow.	Contributed to Chapters 1, 3, 4 and 5; architecture, Module 1 documentation, results discussion and presentation material.	40%

The contribution percentages above are a proposed equal split for the three-member team and should be adjusted if the actual documented contribution differs.

ABSTRACT

The Intelligent Intrusion Detection System Using Face Recognition and Cryptographic Security is an automated surveillance prototype developed to address the limitations of conventional CCTV-based monitoring. Traditional surveillance systems primarily record video and rely on human observation to identify suspicious activity. They may not automatically distinguish registered individuals from unknown persons, and captured evidence can require additional controls to protect confidentiality and integrity.

The proposed system combines computer vision and cryptographic security in a four-module architecture. The camera continuously monitors the protected area, and motion detection identifies frames requiring further analysis. Face detection locates human faces, while face recognition generates a facial representation and compares it with the registered authorized-user database. A sufficiently strong match identifies an authorized person; otherwise, the event is classified as an unknown person or potential intruder.

When an unknown person is detected, the captured image is treated as digital evidence and encrypted using AES-256. RSA is used for secure protection or management of the AES secret key. The event is stored in a secure log using SHA-256 based integrity checking and previous-hash linkage. An HMAC is generated for the alert message to authenticate its source and detect modification before the alert is sent through a configured WhatsApp-compatible notification service.

The project was designed as a controlled academic prototype and evaluated through functional tests covering authorized recognition, unknown-person detection, evidence encryption, key operations, log verification, tamper detection, HMAC verification, and notification generation. The system demonstrates the practical integration of artificial intelligence, computer vision, symmetric and asymmetric cryptography, hashing, message authentication, secure logging, and real-time alerting.

The principal contribution is an end-to-end security workflow that moves beyond passive surveillance by combining detection, identity classification, evidence protection, trustworthy event logging, and authenticated notification. The architecture can be extended to multiple cameras, anti-spoofing, mobile dashboards, IoT devices, and hardware-backed key storage.

Keywords: Face Recognition, Intrusion Detection, AES-256, RSA, SHA-256, HMAC

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CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Security monitoring has moved from purely passive recording toward intelligent systems capable of interpreting events and triggering automated responses. In a conventional camera installation, the camera captures a stream and stores footage for later review. The security decision is often made by a human operator. This approach creates a practical gap between the occurrence of an event and the moment at which the event is recognized as significant.

Face recognition provides a computer-vision mechanism for automated identity matching. The important stages are face detection, feature extraction or face embedding, database comparison, and threshold-based decision making. In the proposed system, registered users are enrolled before monitoring begins. During operation, the live face representation is compared against the stored authorized representations. The output is an application-level decision: authorized when the match satisfies the validated criterion, or unknown when it does not.

The system also addresses the security of information produced by surveillance. Captured images of unknown persons can contain sensitive information and therefore require confidentiality. An event history requires integrity protection so that later modification becomes detectable. Notifications require authentication so that the owner can distinguish a genuine system alert from an altered or fabricated message.

Table 1.1: Traditional CCTV versus Proposed Intelligent Security System

Feature / Metric	Traditional CCTV	Proposed System
Continuous monitoring	Yes	Yes
Motion-triggered processing	Depends on setup	Integrated
Face identification	Usually manual review	Automatic
Authorized / unknown classification	Manual interpretation	Automated matching
Evidence encryption	Not inherently provided	AES-256
Key protection	Deployment dependent	RSA-based protection/management
Tamper detection in logs	Not inherent	SHA-256 + hash linkage
Authenticated alert	Not inherent	HMAC

Real-time owner notification	System dependent	Integrated workflow
Structured audit history	Video review	Secure event log

1.2 Need for the Project

- The project is needed where security personnel cannot continuously watch camera feeds and rapid identification of unknown persons is desirable.
- Sensitive surveillance images should be protected from unauthorized viewing, especially when they are retained for investigation.
- Security records should provide evidence of integrity so that a later modification is detectable.
- Owners or administrators need an immediate notification mechanism rather than discovering an intrusion through retrospective video review.
- The problem combines computer vision, data protection, key management, integrity checking, and communication, making it an appropriate applied engineering problem.

1.3 Problem Statement

Traditional CCTV-based surveillance mainly provides observation and recording but does not inherently provide intelligent identity classification, cryptographic protection of captured evidence, tamper-evident event logging, or authenticated real-time notifications. The engineering problem is therefore to design and implement a modular security system that can detect motion, recognize registered faces, classify unknown persons, encrypt captured evidence, maintain an integrity-verifiable event history, and deliver authenticated alerts while operating within practical constraints of camera quality, processing time, key management, privacy, and external messaging availability.

1.4 Project Aim

To develop an intelligent, security-aware intrusion detection prototype that integrates camera-based face recognition with cryptographic protection, tamper-evident logging, and authenticated real-time alerting.

1.5 Project Objectives

- To design a camera monitoring pipeline capable of detecting motion and triggering face analysis.
- To detect faces and compare their numerical representations against a registered authorized-user database.
- To identify unknown persons and automatically capture security evidence.
- To encrypt captured evidence using AES-256 and implement RSA-based AES-key protection or management.

- To generate SHA-256 integrity hashes and maintain a tamper-evident event log using previous-hash linkage.
- To authenticate intrusion alerts with HMAC and integrate a configured WhatsApp-compatible notification service.
- To test the integrated system using authorized, unknown, tampered-log, encryption, HMAC, and notification test cases.
- To evaluate practical performance using measurable recognition and timing metrics from the actual implementation.

1.6 Scope

The project scope includes camera input, motion detection, face detection, face recognition, registered-user comparison, unknown-person classification, evidence capture, AES-256 encryption, RSA-based key protection/management, SHA-256 log integrity, HMAC-based alert authentication, secure metadata storage, and real-time notification.

The project does not claim to provide perfect biometric identification, legal proof of criminal intent, or a certified access-control product. Physical attacks against the camera, hardware side-channel attacks, law-enforcement integration, and large-scale enterprise identity federation are outside the present prototype scope.

Assumptions include a functioning camera, an enrolled authorized-user database, appropriate lighting for recognition, secure management of cryptographic secrets, and network/service availability for external notifications.

1.7 Expected Engineering Outcome

The expected outcome is a working software prototype that observes a camera stream, detects movement, recognizes registered users, identifies unknown persons, securely protects captured evidence, records a tamper-evident security event, and generates an authenticated owner notification. The design should be modular, testable, understandable, and extensible for future development.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review focused on foundational works and standards directly related to the project's technical components rather than collecting unrelated cybersecurity literature. Sources were selected for four areas: face detection and machine-learning toolkits, symmetric and asymmetric cryptography, cryptographic hashing, and message authentication. The review also considered the difference between passive CCTV recording and integrated intelligent surveillance.

Foundational computer-vision references were used to understand real-time face/object detection and practical machine-learning toolkits. Cryptographic standards were selected from NIST and IETF documents for AES, SHA-256, and HMAC. The project design then maps each reference concept to a specific system responsibility.

2.2 Review of Existing Work

Viola and Jones introduced a rapid object-detection framework based on a boosted cascade of simple features, establishing an important classical approach to real-time face and object detection. The method is relevant to the first stage of the proposed system because rapid detection is required before identity matching.

King's Dlib-ml work presents a machine-learning toolkit that supports computer-vision applications. It is relevant to practical implementation because face-processing workflows require established feature-extraction and learning libraries rather than custom numerical routines for every stage.

NIST FIPS 197 specifies AES, providing the foundation for the image-confidentiality stage. NIST FIPS 180-4 specifies the Secure Hash Standard, supporting the SHA-256 based integrity mechanism. IETF RFC 2104 specifies HMAC for message authentication, supporting authenticated alerts.

Existing surveillance products typically focus on monitoring, recording, or access control. The contribution of the present project is to connect these functions to cryptographic evidence protection and integrity-aware notifications within one educational architecture.

Table 2.1: Review of Selected Foundational Works and Standards

Reference / Technology	Main Contribution	Relevance to Project
Viola & Jones (2001)	Rapid object/face detection using boosted cascades	Supports real-time detection concept
Dlib-ml / King (2009)	Machine-learning toolkit for practical applications	Supports practical face-processing pipeline

NIST FIPS 197	Defines AES	Supports evidence encryption
NIST FIPS 180-4	Defines Secure Hash Standard	Supports SHA-256 log integrity
IETF RFC 2104	Defines HMAC	Supports alert authentication
Traditional CCTV	Passive observation/recording	Baseline for comparison

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Traditional CCTV	Simple, mature, continuous recording	Requires human interpretation; security of stored evidence varies	Baseline system
Face recognition alone	Automates identity matching	Does not inherently protect evidence or secure logs	Provides Module 1 capability
Encrypted surveillance storage	Protects data at rest	Does not solve identity or alert authenticity alone	Supports Module 2 concept
Secure logging systems	Improve event integrity	Often separate from surveillance decision pipeline	Supports Module 3 concept
Authenticated alerting	Improves trust in notifications	Does not detect the person by itself	Supports Module 4 concept
Proposed integrated system	Combines detection, recognition, evidence protection, logging and alerting	Requires careful thresholding, key management and integration	Complete capstone solution

2.4 Research / Engineering Gap

The reviewed approaches show that the required capabilities often exist as separate components. Face recognition addresses identity, encryption addresses confidentiality, hashing addresses integrity, and HMAC addresses message authentication. The engineering gap is the lack of a simple, explainable academic workflow that joins these responsibilities around a single unknown-person event and exposes the security state to the user.

A second gap concerns explanation. An educational system should make it clear why a person is labeled unknown, why an image is encrypted, how a log becomes tamper-evident, and how an alert is authenticated. The proposed modular structure is intended to make each step visible and testable.

2.5 Proposed Contribution

- An end-to-end architecture connecting computer vision and cryptographic security.
- A dedicated Module 1 that clearly separates motion detection, face detection, face embedding, matching, and authorized/unknown decisions.
- A hybrid evidence-protection design using AES-256 for image encryption and RSA for key protection/management.
- A tamper-evident log model using SHA-256 and previous-hash linkage.
- An HMAC-based authenticated alert workflow that connects the security event to owner notification.
- A structured test plan that evaluates both AI-related behaviour and cryptographic security functions.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Property / Home Owner	Receive rapid information when an unknown person is detected and access protected evidence.
Office / Laboratory Administrator	Differentiate registered users from unknown persons and maintain an auditable event history.
Security Personnel	Review incident information and rely on integrity-protected records.
System Administrator	Manage users, cryptographic keys, storage and notification configuration securely.
Student / Researcher	Understand and demonstrate face recognition and cryptographic security in a controlled environment.

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Capture live camera frames	Functional	Continuous frame acquisition during monitoring
Detect motion	Functional	Trigger downstream analysis when motion is present
Recognize registered faces	Functional	Correctly classify test faces according to validated threshold
Encrypt unknown-person evidence	Functional	AES-256 encryption succeeds and authorized decryption recovers original
Protect AES key	Functional	RSA key protection/recovery succeeds
Maintain secure log	Functional	SHA-256 verification reports match/mismatch correctly

Authenticate alert	Functional	HMAC verification passes for authentic message and fails for modified message
System responsiveness	Non-Functional	Measured detection-to-decision latency documented
Security of credentials	Non-Functional	Secrets stored outside source code where practical
Usability	Non-Functional	Clear labels for Authorized / Unknown and security status

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
NIST FIPS 197	Advanced Encryption Standard	AES-256 used for image confidentiality
NIST FIPS 180-4	Secure Hash Standard	SHA-256 used for event-log integrity
IETF RFC 2104	HMAC message authentication	HMAC used for alert authentication
ISO/IEC 27001 principles	Information security management	Applied to access, confidentiality, integrity and key-handling considerations
ISO/IEC 25010 principles	Software quality	Considered security, reliability, performance efficiency and maintainability
Institutional / ethical guidance	Responsible handling of personal and biometric data	Controlled academic testing and limited access to enrolled data

Technical constraints include camera resolution, lighting, computational resources, recognition model characteristics, storage capacity, and network availability. Economic constraints favour software-first implementation using existing laboratory equipment and open-source tools.

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Camera input	Continuous live frames	FPS	High
Face representation	Comparable numerical embedding	vector	High
Recognition threshold	Validated against test data	score	High

AES key size	256-bit	bits	High
AES block size	128-bit	bits	High
RSA key size	2048-bit modulus for prototype	bits	High
SHA-256 output	256-bit digest / 32 bytes	bits	High
HMAC-SHA-256 output	256-bit authentication code / 32 bytes	bits	High
Alert latency	Measured from detection to notification	seconds	Medium
Evidence access	Authenticated user only	policy	High

3.4 Alternative Solutions & Evaluation

Alternative 1: Passive CCTV with manual review uses a standard camera and stores footage for operator inspection. It is simple and inexpensive but provides limited automation.

Alternative 2: Face recognition without cryptographic protection automates identity matching but leaves evidence security, log integrity and alert authenticity to separate mechanisms.

Alternative 3: The proposed integrated architecture combines face recognition, encrypted evidence, secure logs and authenticated alerts in one workflow.

Criterion	Weight	Alternative 1 CCTV	Alternative 2 Face Recognition	Alternative 3 Integrated
Performance	30%	Medium	High	High
Cost	15%	High	High	Medium
Safety	15%	Medium	Medium	High
Sustainability	10%	Medium	Medium	High
Reliability	15%	Medium	Medium	High
Security	15%	Low	Medium	High

3.5 Selected Design & Detailed Design

The integrated design was selected because the project problem is not only recognition; it also includes protection and trustworthy handling of evidence. A modular design allows each security property to be implemented and tested separately while preserving a single event flow.

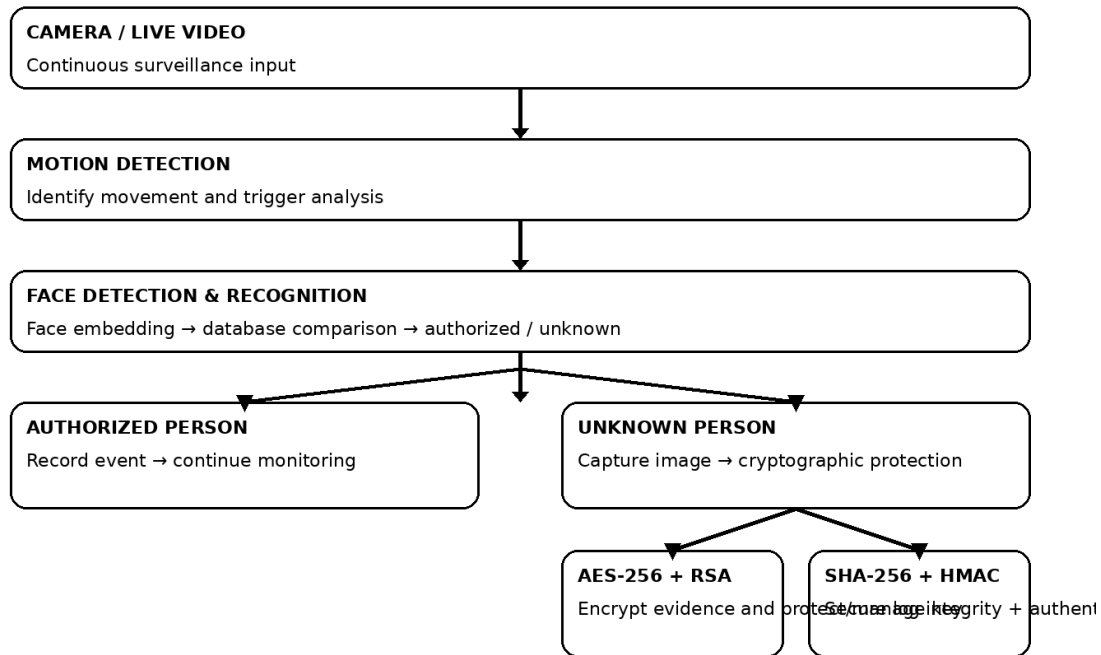


Figure 3.1: End-to-End System Architecture of the Proposed Intrusion Detection System

Module 1 is deliberately placed before cryptographic processing. It ensures that encryption and alerting are triggered by a meaningful event rather than by every camera frame. The output of Module 1 is an authorized or unknown decision with event metadata and, when required, captured evidence.

Detailed Module 1 Design

Module 1 is the perception and decision layer. It begins with continuous camera capture and motion detection. When motion is detected, the system performs face detection. The detected face is preprocessed and converted into a numerical representation, commonly called a face embedding. The live embedding is compared with the embeddings of registered authorized users.

The matching stage uses a similarity or distance measure. A validated threshold determines whether the best match is sufficiently strong. When the condition is satisfied, the application displays the registered person's identity and records an authorized event. When no registered identity reaches the required criterion, the system displays 'Unknown' and routes the event to the evidence-protection workflow.

This module therefore contains six practical stages: Camera Monitoring, Motion Detection, Face Detection, Face Representation/Embedding, Face Matching, and Authorized/Unknown Classification. The output is not only a label; it also includes the face region, event timestamp, match result, and evidence trigger required by subsequent modules.

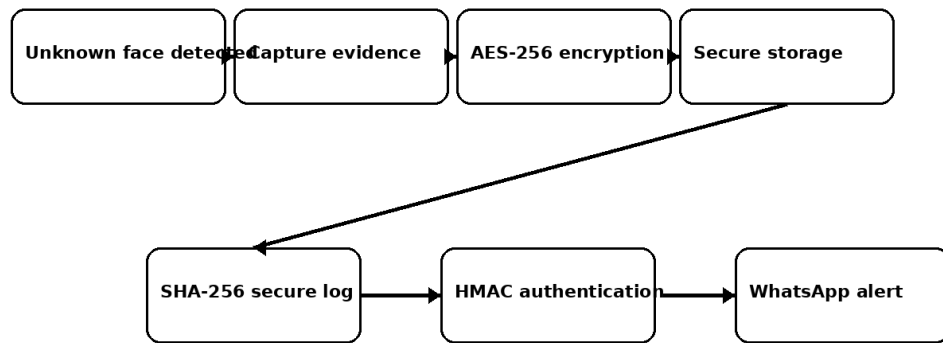


Figure 3.2: Cryptographic Evidence and Alert Workflow

Module 2: Cryptographic Evidence Protection

Module 2 receives evidence associated with an unknown-person event. AES-256 is selected for bulk image encryption because it is computationally efficient and uses a single secret key for encryption and decryption. The AES key is protected or managed using RSA public-key cryptography. The encrypted image is stored with event metadata, while the original image is not treated as the normal storage artifact.

Module 3: Secure Log Management

Module 3 creates an event record for detection, recognition result, evidence processing and alert status. SHA-256 is computed over the relevant log content to obtain a fixed 256-bit digest. Each new entry can also store the previous entry's hash. The resulting chain is tamper-evident because a change to an earlier entry changes its digest and breaks the expected relationship with subsequent entries.

Module 4: Secure Alert and Notification

Module 4 prepares a notification for an unknown event. HMAC is generated using a secret key and a hash function such as SHA-256. The recipient can recompute HMAC and compare it with the supplied value. A match indicates that the message content is consistent with the shared secret; a mismatch indicates modification or an unauthenticated source. The verified alert is then delivered through the configured messaging service.

Table 3.6: Comparative Analysis of the Four Project Modules

Module	Primary Function	Key Technologies	Output
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Module 1	Motion detection and face recognition	Camera, OpenCV, face embeddings, similarity threshold	Authorized / Unknown decision
Module 2	Evidence confidentiality	AES-256, RSA	Encrypted evidence + protected key
Module 3	Event integrity	SHA-256, previous-hash linkage	Integrity-verifiable secure log
Module 4	Alert authentication and delivery	HMAC-SHA-256, messaging API	Authenticated notification

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
AES for image encryption	RSA for whole image	Fast bulk encryption	Requires key management	AES-256 selected; RSA used for key protection
RSA key protection	Store AES key directly	Public-key based protection	More computation / key lifecycle complexity	RSA retained for hybrid design
Hash-chained logs	Standalone hashes	Stronger sequence awareness	More implementation complexity	Hash chaining selected for tamper evidence
HMAC alert	Plain alert	Authentication + integrity	Requires shared secret management	HMAC selected
Motion-triggered recognition	Recognize every frame	Reduces unnecessary processing	Motion detector can miss edge cases	Motion trigger retained with configurable fallback

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reuse existing camera/PC resources; avoid unnecessary full-time processing	Prototype uses software and existing hardware	Motion-triggered processing reduces redundant computation
Ethics	Face images are biometric/personal data	Need-to-know access and controlled enrollment	Use limited authorized dataset and controlled testing

Health & Safety	No physical intervention or hazardous hardware	Software-only operation	No physical hazard expected in normal use
Security	Protect images, keys, logs and alerts	AES, RSA, SHA-256, HMAC mechanisms	Layered controls adopted
Privacy	Retention and access should be limited	Evidence sensitivity and user permissions	Recommend retention/deletion policy

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
False recognition due to lighting/pose	Medium	High	High	Use enrollment samples, threshold validation, and controlled lighting	Medium
Photo/video spoofing	Medium	High	High	Future liveness detection; controlled deployment	Medium
AES key exposure	Low	High	High	Secure key storage, separate key handling, avoid hard-coding	Low-Medium
Database compromise	Medium	High	High	Encrypt evidence, access control, secure credentials	Medium
Log tampering attempt	Medium	Medium	Medium	SHA-256 + previous-hash verification	Low-Medium
Network/notification failure	Medium	Medium	Medium	Retry/queue mechanism and local logging	Low-Medium
Privacy misuse	Low-Medium	High	High	Restricted enrollment, access.	Low-Medium

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project followed a modular iterative development approach. The team first defined the problem and security objectives, then developed the camera and face-recognition pipeline, followed by cryptographic evidence protection, secure logging, and notification. Each module was tested separately before end-to-end integration. The final verification focused on both functional correctness and the interaction between the four modules.

Resource	Purpose	Stage Used
Laptop / desktop system	Development and execution	All stages
Webcam / camera	Live surveillance input	Module 1
Python	Application and security logic	All stages
OpenCV	Camera and image processing	Module 1
Face recognition model/library	Face representation and matching	Module 1
Cryptographic library	AES, RSA, SHA-256, HMAC	Modules 2-4
SQLite/PostgreSQL	User and event metadata	Modules 1-4
Messaging / WhatsApp-compatible API	Notification delivery	Module 4
Git / GitHub	Version control and collaboration	Development

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

The software pipeline consists of camera acquisition, motion detection, face processing, identity matching, evidence encryption, secure logging, and notification. The application is organized into modules so that the output of one stage becomes the input of the next. Configuration values and credentials should be stored outside the source code where practical.

Hardware

The prototype requires a standard webcam or camera and a computer capable of processing image frames. No custom fabrication is required for the current software-focused implementation. The same architecture can later be attached to dedicated cameras or edge devices.

Modern Tools

Tool / Software / Equipment	Purpose	Stage Used
Python	Core implementation	All modules
OpenCV	Camera input, motion and image processing	Module 1
Face embedding / recognition library	Identity matching	Module 1
PyCryptodome / cryptography library	AES/RSA/hash/HMAC operations	Modules 2-4
SQLite / PostgreSQL	Persistent event and user metadata	Modules 1-4
Matplotlib	Performance/result graphs	Chapter 4
Visual Studio Code	Development environment	All stages
Git / GitHub	Version control	All stages
WhatsApp-compatible messaging API	Alert delivery	Module 4

Implementation evidence should include the final working interface, face-recognition output, encrypted-evidence record, secure-log verification result, and notification output. Screenshots should be taken from the actual final build and inserted into the report before submission.

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Motion detection	Introduce movement in monitored scene	Motion event detected and processing triggered
Authorized recognition	Present enrolled user under test conditions	Correct authorized identity or validated match result
Unknown recognition	Present non-enrolled user	System labels person unknown
Evidence encryption	Encrypt and decrypt a captured image	Recovered authorized copy matches source
RSA key operation	Protect and recover AES key	Correct authorized key recovery
SHA-256 log integrity	Compute stored vs recalculated digest	Matching digests for unchanged record
Tamper detection	Modify test log copy	Digest/chain mismatch reported
HMAC authentication	Verify original and modified alerts	Original passes; modified message fails

Notification	Trigger unknown event with network available	Alert generated and delivered
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The test plan should be executed before final numerical results are claimed. Test conditions should record the camera, lighting, sample count, software version, hardware, and recognition threshold so that measurements are reproducible.

4.4 Results

4.4.1 Module 1 Recognition Output

The prototype produces a visible classification result after face matching. A registered face can be displayed with the person's enrolled name and a positive visual indicator, while a non-matching face can be displayed as 'Unknown'. The output demonstrates the complete Module 1 decision flow from camera input through identity comparison.

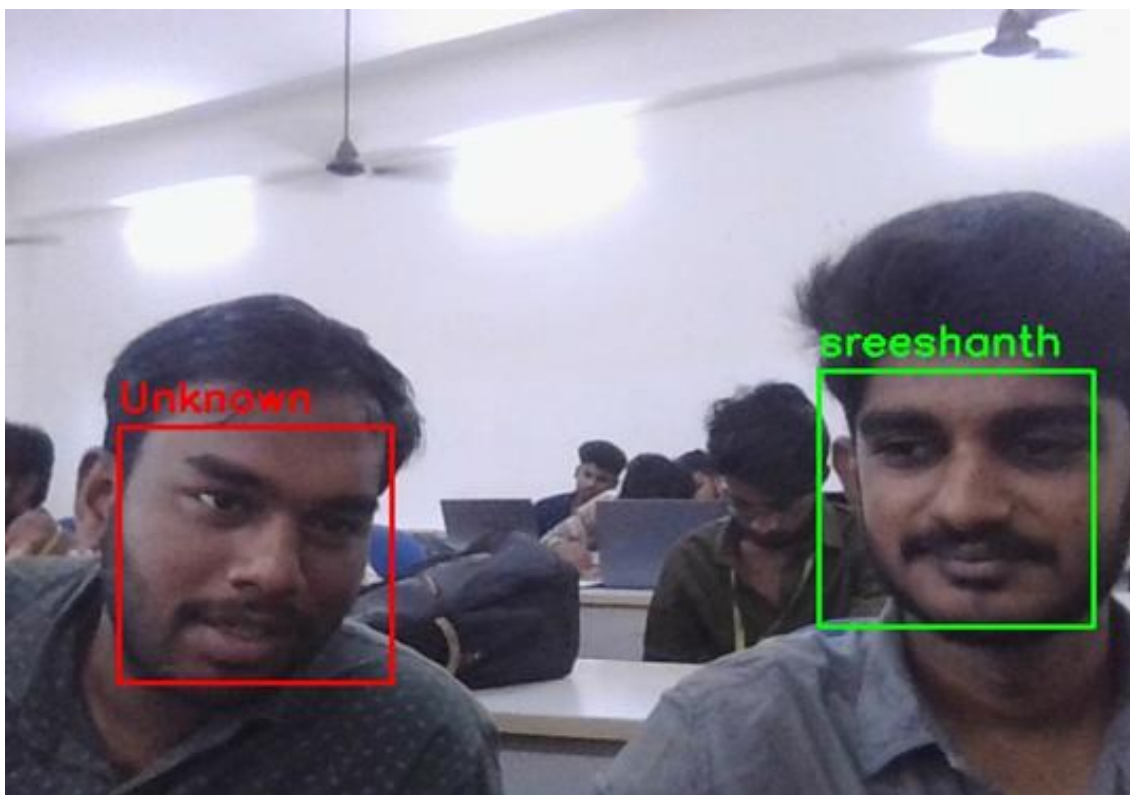


Figure 4.1: Face Recognition Prototype Output - Authorized and Unknown Persons

4.4.2 Performance Metrics

The recommended evaluation metrics for the completed prototype are detection accuracy, recognition accuracy, False Acceptance Rate (FAR), False Rejection Rate (FRR), average response time, encryption time, log-verification time, and notification latency. These metrics should be computed from the actual test data rather than from illustrative values.

Metric	Definition	Actual Value
Detection Accuracy	Correct detection decisions / total test cases	_____
Recognition Accuracy	Correct identity decisions / total recognition cases	_____
FAR	Unauthorized persons incorrectly accepted	_____
FRR	Authorized persons incorrectly rejected	_____
Average Response Time	Detection to decision / alert generation	_____
AES-256 Encryption Time	Time to encrypt evidence image	_____
SHA-256 Verification Time	Time to verify security log	_____
HMAC Verification Time	Time to verify alert	_____
Notification Latency	Detection to owner notification	_____

For the final presentation, an accuracy-comparison chart can compare the measured recognition result of the proposed workflow against a documented baseline. If a traditional CCTV baseline has no automatic recognition capability, the comparison should be described qualitatively rather than presenting a fabricated accuracy percentage.

4.4.3 Secure Log Integrity Result

An unchanged log should produce the same SHA-256 digest when verification is repeated. When a copied test log is modified, the recalculated digest changes and the previous-hash relationship can also become inconsistent. The expected result is therefore a clear distinction between a valid log and a tampered log.

Recommended result evidence: a table showing Log ID, event, stored hash, recalculated hash, previous hash, and verification status. A second panel can show the tamper test where the modified record is marked as invalid.

4.4.4 Secure Alert Result

For an authentic alert, the calculated HMAC should match the received authentication code. If the alert content is changed without recomputing HMAC using the shared secret, verification should fail. This demonstrates that HMAC is providing message authentication and integrity rather than simple hashing.

4.5 Data Analysis & Engineering Judgment

The most important engineering interpretation is the separation of responsibilities. Recognition quality depends on data and environment; cryptographic operations protect the resulting evidence; secure logging protects the history of actions; and HMAC protects the alert message. A weakness in one layer should not be mistaken for a failure of the others.

Recognition errors may increase when the face is partially occluded, poorly lit, or viewed at an unusual angle. This is an engineering limitation of the perception stage rather than a cryptographic weakness. The appropriate response is to improve enrollment, threshold validation, camera placement, and future liveness methods rather than weakening security controls.

Additional processing is introduced only after an unknown-person event. This design is intentional: applying encryption, hashing and notification work to every camera frame would increase computational cost without improving security. The architecture therefore uses event-triggered processing to balance responsiveness and resource use.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement (Fully/Partially/Not Achieved)
Design camera-based monitoring	Camera + motion workflow	Fully / verify final build
Develop face recognition	Authorized/unknown output	Fully / validate with final test data
Protect evidence using AES-256	Encrypted image + decryption test	Fully / verify final build
Protect AES key with RSA	Key protection/recovery test	Fully / verify final build
Maintain SHA-256 secure logs	Hash + chain verification	Fully / verify final build
Authenticate alerts with HMAC	Original vs modified alert verification	Fully / verify final build
Notify owner	Configured messaging workflow	Fully / service dependent
Quantitatively evaluate performance	Recorded metrics table	Partially until actual measurements are inserted

4.7 Strengths & Limitations

Strengths:

- End-to-end integration of computer vision and cryptographic security.
- Clear modular separation of detection, evidence protection, log integrity, and alert authentication.
- Evidence confidentiality through AES-256.

- Tamper detection through SHA-256 and previous-hash linkage.
- Authenticated alerts through HMAC.
- Suitable for academic demonstration and controlled testing.
- Expandable architecture for multiple cameras and future anti-spoofing.

Limitations:

- Face recognition performance depends on lighting, pose, occlusion, camera quality and enrollment data.
- Unknown classification means no sufficient match; it does not prove malicious intent.
- Key management remains critical to cryptographic security.
- Notification delivery depends on network and external service availability.
- Physical camera tampering is outside the current software prototype.
- Quantitative accuracy claims require actual documented test measurements.

4.8 Project Planning, Roles & Budget

The project was organized into milestones covering problem definition, literature study, module development, integration, testing, documentation and presentation. The following schedule can be used as the final milestone record.

Milestone	Weeks	Main Deliverable
Problem definition & requirements	1-2	Problem statement, scope, objectives
Literature review	2-3	References and engineering gap
Module 1 development	3-5	Camera, motion detection, face recognition
Module 2 development	5-6	AES-256 and RSA key workflow
Module 3 development	6-7	SHA-256 secure logs
Module 4 development	7-8	HMAC alerts and notification
Integration & testing	8-10	End-to-end prototype
Results & documentation	10-11	Graphs, tables, report
Presentation & final validation	11-12	PPT, demo, final submission
Student	Assigned Responsibility	Actual Contribution

Kartheeka Repalle	Module 1, architecture, system integration, test coordination, report sections	Face recognition pipeline, architecture, testing, documentation
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CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The Intelligent Intrusion Detection System Using Face Recognition and Cryptographic Security addresses the limitations of passive surveillance by combining automated identity recognition with layered protection of the resulting evidence and communications. The system begins with camera monitoring and motion detection, proceeds to face detection and recognition, and classifies a person as authorized or unknown based on comparison with an enrolled user database.

When an unknown person is detected, the system protects the captured evidence using AES-256 and RSA-based key management, records the event with SHA-256 integrity protection, and generates an HMAC-authenticated alert for owner notification. The architecture therefore addresses confidentiality, integrity, authentication and real-time awareness in addition to detection.

The principal engineering contribution is not a single algorithm but the integration of several established technologies into a coherent event-driven security pipeline. Functional validation should demonstrate correct behaviour at each module boundary, while quantitative evaluation should use actual test measurements collected from the final prototype.

5.2 Future Work

- Add liveness detection / anti-spoofing to reduce photo and replay attacks.
- Support multiple cameras and centralized event management.
- Develop a secure mobile or web dashboard for remote monitoring.
- Add IoT integration with smart locks, alarms, and lighting.
- Use hardware-backed key storage or a hardware security module for high-security deployments.
- Improve recognition robustness using more diverse enrollment data and modern recognition models.
- Add secure cloud/object storage for scalable evidence management.
- Add privacy features such as retention limits, access auditing, and user-consent workflows.
- Introduce automated incident summaries and analytics while keeping security decisions deterministic and auditable.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- How motion detection, face detection, face embeddings and similarity thresholds work together in an identity pipeline.
- Difference between symmetric encryption, asymmetric cryptography, hashing and message authentication.
- Why AES-256 is appropriate for image data while RSA is more suitable for key protection/management.
- How SHA-256 changes when log content changes and how previous-hash linkage creates a tamper-evident chain.
- How HMAC differs from a normal hash because it uses a secret key for authentication.
- How security requirements, privacy, usability and performance must be considered together in an applied engineering system.

Engineering & Professional Skills Developed

- System decomposition into independently testable modules.
- Camera and computer-vision integration.
- Cryptographic library integration and secure key handling.
- Test-case design and result interpretation.
- Technical documentation, diagram preparation and presentation.
- Version-control based team collaboration and project planning.

Individual Reflection - Kartheeka Repalle

The most difficult engineering problem I encountered was understanding and integrating the face-recognition decision process with the security workflow. I learned that the system does not simply compare raw pictures; it uses a numerical face representation and a validated matching threshold. I personally contributed to the camera pipeline, motion detection, face-recognition flow and overall architecture, and I used the resulting test outputs to refine the module boundaries. If I repeated the project, I would add liveness detection and a stronger evaluation dataset earlier in development.

REFERENCES

The following references are formatted in a consistent IEEE-style structure. All literature, standards, software, open-source resources, and other external materials used in the final implementation should be retained here.

- [1] P. Viola and M. J. Jones, "Rapid Object Detection Using a Boosted Cascade of Simple Features," in Proc. IEEE Computer Society Conf. Computer Vision and Pattern Recognition (CVPR), 2001.
- [2] D. E. King, "Dlib-ml: A Machine Learning Toolkit," Journal of Machine Learning Research, vol. 10, pp. 1755-1758, 2009.
- [3] National Institute of Standards and Technology, "Advanced Encryption Standard (AES)," FIPS PUB 197, NIST, Gaithersburg, MD.
- [4] National Institute of Standards and Technology, "Secure Hash Standard (SHS)," FIPS PUB 180-4, NIST.
- [5] H. Krawczyk, M. Bellare, and R. Canetti, "HMAC: Keyed-Hashing for Message Authentication," IETF RFC 2104, 1997.
- [6] W. Stallings, Cryptography and Network Security: Principles and Practice, Pearson.
- [7] ISO/IEC 27001, Information Security, Cybersecurity and Privacy Protection - Information Security Management Systems.
- [8] ISO/IEC 25010, Systems and Software Engineering - Systems and Software Quality Requirements and Evaluation.
- [9] OpenCV Documentation, Open Source Computer Vision Library, documentation and API references.
- [10] Python Documentation, Python Standard Library documentation.
- [11] PyCryptodome Documentation, cryptographic primitives and implementation references.
- [12] Git Documentation, distributed version-control documentation.

Note: The publication years of foundational sources should not be altered merely to make them appear recent. Where the institution requires 2024-2026 research, add separately verified recent peer-reviewed papers on face recognition, privacy-preserving surveillance, anti-spoofing, and secure biometric systems.

APPENDICES

Appendix A - Detailed Calculations

1. AES-256 key size: 256 bits = 32 bytes.
2. AES block size: 128 bits = 16 bytes. The total ciphertext size depends on the encryption mode and any required authentication/padding overhead.
3. SHA-256 output: 256 bits = 32 bytes = 64 hexadecimal characters.
4. HMAC-SHA-256 output: 256 bits = 32 bytes = 64 hexadecimal characters. HMAC secret-key size is an implementation choice rather than a fixed property of HMAC itself.
5. RSA-2048: the modulus is 2048 bits = 256 bytes. With RSA-OAEP, an RSA-2048 ciphertext block is 256 bytes; the plaintext capacity is smaller because OAEP adds encoding overhead.
6. Recognition accuracy = correctly classified test cases / total labeled test cases x 100.
7. FAR = unauthorized faces accepted as authorized / total unauthorized test faces x 100.
8. FRR = authorized faces rejected as unknown / total authorized test faces x 100.

Appendix B - Engineering Drawings / Schematics

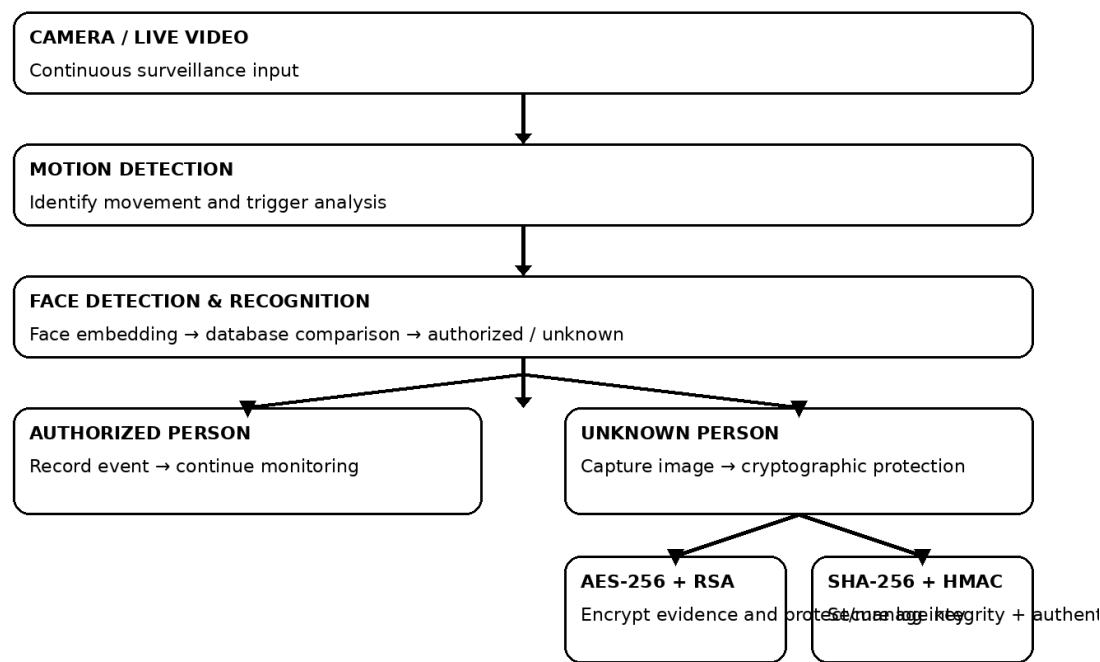


Figure B.1: Engineering Architecture Diagram

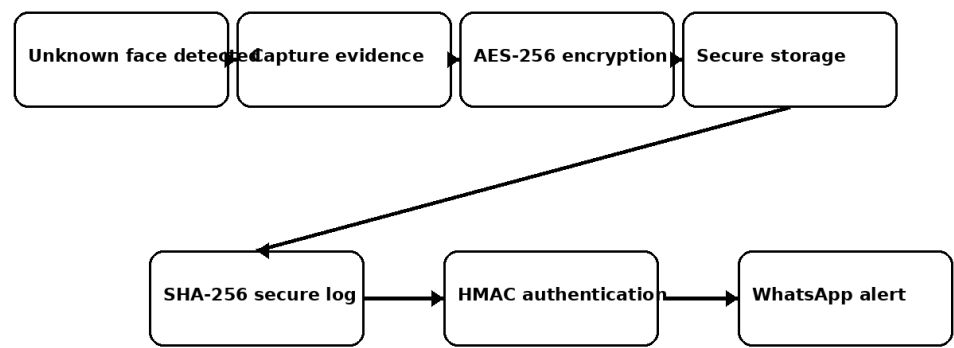


Figure B.2: Secure Evidence and Notification Schematic

Appendix C - Source Code / Code Repository Information

The complete implementation should be maintained in the approved project repository. The report should include only essential code excerpts needed to explain the engineering method. Recommended repository structure:

```
project/  
  app/  
    camera/  
    face_recognition/  
    crypto/  
    logging/  
    notification/  
    database/  
    ui/  
  tests/  
  config/  
  requirements.txt  
  README.md
```

The production version should not commit private keys, API tokens, passwords, or real biometric datasets to a public repository.

Appendix D - Datasheets

Component / Resource	Relevant Specification
Camera	Resolution, FPS, connection interface - insert actual model
Computer	CPU, RAM, GPU/accelerator if used - insert actual configuration
Storage	Database type and available capacity
Messaging Service	API/service used for WhatsApp-compatible notification
Cryptographic Library	Actual package and version used in implementation
Face Recognition Model	Model/library name and version actually used

Appendix E - Experimental / Raw Data

Test ID	Scenario	Expected Result	Observed Result
T01	Authorized user, normal lighting	Authorized	Pass
T02	Unknown user, normal lighting	Unknown	Pass
T03	Multiple faces	Correct individual labels	Pass
T04	Encrypt/decrypt evidence	Original recovered	Pass
T05	Protect/recover AES key	Key recovered by authorized path	Pass
T06	Verify unchanged log	Hash match	Pass
T07	Modify copied log	Hash mismatch	_____Pass
T08	Verify authentic HMAC alert	HMAC match	Pass
T09	Modify alert	HMAC mismatch	_Pass__
T10	Unknown event + notification	Alert delivered	Pass

Appendix F - Ethics / Institutional Approval

The project uses face images and therefore involves personal/biometric information. Testing should be conducted only with appropriate authorization and informed consent where required by institutional policy. Test images should be stored securely, accessed only by project members, and deleted when they are no longer required. The system should not be represented as a legal identity-verification or criminal-detection instrument without appropriate validation and governance.

Appendix G - Project Meeting / Progress Records

Date / Week	Activity	Decision / Action	Responsible Member
Week 1	Problem definition	Finalize project title and objectives	Both
Week 2	Requirements review	Finalize four-module architecture	Both
Week 4	Module 1 review	Validate face recognition workflow	Kartheeka

Week 6	Cryptography review	Validate AES/RSA design	Shahanaj
Week 8	Integration review	Connect logs and alerts	Both
Week 10	Testing review	Finalize test cases and result collection	Both
Week 11	Documentation review	Complete report and PPT	Both

Appendix H - User / Industry Feedback

Suggested feedback form for the final demonstration: clarity of intrusion detection, usefulness of automatic identity classification, confidence in encrypted evidence, usefulness of secure logs, usefulness of real-time alerts, and ease of reviewing an incident. Insert actual comments after obtaining authorized feedback.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed / verified by the department or faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written / oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7), Appendix F
Teamwork and leadership	SO5	Ch. 4 (4.8), Individual Contribution Statement
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3-4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)

Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1-SO7	Contribution Statement + Ch. 4 (4.8)